Task 6.1P

Module Summary

Summary

This module/task covered how the network layer sends IP datagrams between different networks. Internet uses a best effort service, with no gaurantees for delivery, the order, bandwidth or timing. Forwarding moves a packet from a router input to the correct output and routing decides it’s path. A router has input ports, a switching fabric, output ports and a routing processor. Buffer queues can add delay or cause packet loss.

The CloudDeakin notes explained the router’s input/output ports, switching fabric and the routing processor. Input ports use packet headers and forwarding tables before switching. When arrival rates are too high, queues may case delays, buffer overflows and/or packet loss. Head of Line blocking can delay other packets. This connects to L4S, where Active Queue Management uses ECN to mark packet congestion before queues become full, allowing endpoints to respond quickly. By keeping queues short like this, L4S will provide lower latency and lower packet loss while still managing to maintain a very salable throughput.

While doing the tasks, I reviewed IPv4 fields including it’s length, protocol, TTL, checksum, addresses, identification, flags and fragment offsets. Fragmentation lets IPv4 cross links with smaller MTUs, while the receiver rebuilds the original datagram on it’s end. CIDR seperates the network and host ports. A /26 CIDR suffix has 62 usable addresses, while /23 supports 510 usable hosts. DHCP assigns and renews IP addresses through using a client-server kind of architecture. NAT lets multiple devices in a private network share a public IPv4 address by translating addresses and ports.

In Cisco Packet Tracer, I built the 10.1.1.0/24 and 192.168.1.0/24 LANs, I configured the router, PCs and gateways and tested them with a simple ping. Simulation mode showed the ICMP request and reply in action as intended. I also learned that the network layer is processed three times and the data link layer six times because the router handles incoming and outgoing frames separately.In the notes, IPv6 was also covered, with 128 bit addresses, a fixed 40 byte header, flow label, Next Header, Hop Limit and no header checksum.

Reflection

I was already familar with most of the things covered in this task because I had already gone through the contents of the Module 1 of CCNA study materials and has spent a lot of time using Cisco Packet Tracer. Configuring the router, PCs, IP addresses and gateways was mostly revision for me. The simulation was still useful because it showed how packets were routed in more detail, specially how the routers change the ethernet frame. I also corrected my misunderstanding about the data link layer being processed twice at the router. The queueing content connected well with my L4S and AQM work, so that part was relevant.